

Math 6601 Midterm

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Problem 1.

Suppose that $G : \mathbf{R}^n \rightarrow \mathbf{R}^n$ is a contraction with contraction factor $\alpha \in (0, 1)$ on a closed set $D_0 \subseteq \mathbf{R}^n$ and that $G(D_0) \subseteq D_0$. Then

1. G has a unique fixed point $x^* \in D_0$;
2. for any $x^{(0)} \in D_0$, the sequence $\{x^{(k)}\}$ defined recursively by $x^{(k+1)} = G(x^{(k)})$ converges to x^* ;
3. for any norm $\|\cdot\|$ on $\mathbf{R}^n$, the sequence $\{x^{(k)}\}$ in 2. satisfies the inequalities

$$\|x^{(k)} - x^*\| \leq \frac{\alpha}{1 - \alpha} \|x^{(k)} - x^{(k-1)}\|, \quad k = 1, 2, \dots, \quad (1)$$

and

$$\|x^{(k)} - x^*\| \leq \frac{\alpha^k}{1 - \alpha} \|G(x^{(0)}) - x^{(0)}\|. \quad (2)$$

Proof. Choose any point $x^{(0)}$ and consider the sequence $\{x^{(k)}\}$ from 2. Given integer $m < n$, we can apply the contraction condition m times to show that this sequence satisfies

$$\begin{aligned} \|x^{(n)} - x^{(m)}\| &\leq \alpha^m \|x^{(n-m)} - x^{(0)}\| \\ &\leq \alpha^m \|x^{(n-m)} - x^{(n-m-1)} + x^{(n-m-1)} - \dots + x^{(2)} - x^{(1)} + x^{(1)} - x^{(0)}\| \\ &\leq \alpha^m [\|x^{(n-m)} - x^{(n-m-1)}\| + \dots + \|x^{(1)} - x^{(0)}\|]. \end{aligned}$$

Applying the contraction condition as many times as possible to each term yields

$$\begin{aligned} \|x^{(n)} - x^{(m)}\| &\leq \alpha^m [\alpha^{n-m-1} + \dots + \alpha + 1] \|x^{(1)} - x^{(0)}\| \\ &\leq \alpha^m \|x^{(1)} - x^{(0)}\| \sum_{k=0}^{\infty} \alpha^k \\ &\leq \frac{\alpha^m}{1 - \alpha} \|x^{(1)} - x^{(0)}\|. \end{aligned} \quad (3)$$

Since the right-hand side of (3) converges to 0 as $m \rightarrow \infty$, it follows that $\{x^{(k)}\}$ is Cauchy. By the equivalence of norms on $\mathbf{R}^n$, the subset D_0 is closed with respect to $\|\cdot\|$, and $\mathbf{R}^n$ is complete with respect to $\|\cdot\|$, so D_0 is also complete with respect to $\|\cdot\|$. By applying the condition $G(D_0) \subseteq D_0$ repeatedly, we easily see that $x^{(k)} \in D_0$ for all k . Thus, the sequence $\{x^{(k)}\}$ converges to a point $x^* \in D_0$, which we now show is the unique fixed point of G in D_0 .

Firstly, we show that x^* is a fixed point of G . Let $\varepsilon > 0$ be given. Then we can choose K such $k > K$ implies that $\|x^* - x^{(k)}\| \leq \frac{\varepsilon}{2}$. It follows that

$$\begin{aligned} \|G(x^*) - x^*\| &\leq \|G(x^*) - x^{(K+2)}\| + \|x^* - x^{(K+2)}\| \\ &\leq \|G(x^*) - G(x^{(K+1)})\| + \frac{\varepsilon}{2} \\ &\leq \alpha \|x^* - x^{(K+1)}\| + \frac{\varepsilon}{2} \\ &\leq \alpha \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \leq \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, it follows that $\|G(x^*) - x^*\| = 0$, which implies that $G(x^*) = x^*$; that is, x^* is a fixed point of G .

Secondly, we show that x^* is the only fixed point of G in D_0 . Suppose that $y^* \in D_0$ is a fixed point of G . Then

$$\|x^* - y^*\| = \|G(x^*) - G(y^*)\| \leq \alpha \|x^* - y^*\| \implies (1 - \alpha) \|x^* - y^*\| \leq 0,$$

which implies that $\|x^* - y^*\| = 0$ because $1 - \alpha > 0$, so $x^* = y^*$; that is, x^* is the unique fixed point of G in D_0 . This concludes the proof of 1.

The proof of 2. is now trivial, as the point $x^{(0)}$ chosen in the proof of 1. was arbitrary.

Lastly, to prove 3. we observe that, by the contraction condition,

$$\|x^{(k)} - x^*\| \leq \|x^{(k)} - x^{(k+1)}\| + \|x^{(k+1)} - x^*\| \leq \alpha \|x^{(k)} - x^{(k-1)}\| + \alpha \|x^{(k)} - x^*\|,$$

which clearly implies (1). Furthermore, given $\varepsilon > 0$, we can choose $m > k$ large enough that $\|x^{(m)} - x^*\| \leq \varepsilon$. Then, by (3), we have

$$\|x^{(k)} - x^*\| \leq \|x^{(k)} - x^{(m)}\| + \|x^{(m)} - x^*\| \leq \frac{\alpha^k}{1 - \alpha} \|x^{(1)} - x^{(0)}\| + \varepsilon,$$

which implies (2) because $\varepsilon > 0$ was arbitrary and $x^{(1)} = G(x^{(0)})$. □

Problem 2.

Let $D_0 \subseteq \mathbf{R}^n$ be convex, and let the component functions $g_1, \dots, g_n$ of G be continuously differentiable with bounded derivatives on D_0 . Then G satisfies the Lipschitz condition

$$\|G(x) - G(y)\| \leq L \|x - y\| \quad x, y \in D_0, \quad (4)$$

where $L = \sup_{x \in D_0} \|DG(x)\|$, where DG is the Jacobian matrix function of G , and $\|\cdot\|$ as a matrix norm is the norm induced by $\|\cdot\|$ as a norm on $\mathbf{R}^n$. Note that if $L < 1$, then G is a contraction on D_0 by definition.

Proof. Let $x, y \in D_0$ be given. Then the line segment $\{tx + (1 - t)y : t \in [0, 1]\} \subseteq D_0$ because D_0 is convex, so we may define $h_i : [0, 1] \rightarrow \mathbf{R}$ by $h_i(t) = g_i(tx + (1 - t)y)$, and by the chain rule

$$h'_i(t) = \sum_{j=1}^n \partial_j g_i(tx + (1 - t)y) \cdot (x_j - y_j) = DG_i(tx + (1 - t)y)(x - y),$$

where ∂_j means the partial derivative with respect to the j th component, and DG_i is the i th row of DG . By the fundamental theorem of calculus, we have

$$g_i(x) - g_i(y) = h_i(1) - h_i(0) = \int_0^1 h'_i(t) dt = \int_0^1 DG_i(tx + (1 - t)y)(x - y) dt.$$

It follows that

$$G(x) - G(y) = \int_0^1 \begin{bmatrix} DG_1(tx + (1-t)y)(x-y) \\ \vdots \\ DG_n(tx + (1-t)y)(x-y) \end{bmatrix} dt = \int_0^1 DG(tx + (1-t)y)(x-y) dt.$$

□

Thus,

$$\|G(x) - G(y)\| \leq \int_0^1 \|DG(tx + (1-t)y)(x-y)\| dt \leq \int_0^1 L \|x - y\| dt = L \|x - y\|,$$

if DG is bounded with respect to the induced norm $\|\cdot\|$. This is, indeed, the case, because each component of DG is bounded by hypothesis, so that there exists a $n \times n$ matrix $M = \{m_{ij}\}$ such that $m_{ij} > |DG_{ij}(x)|$ for all $x \in D_0$, where $DG_{ij} = \partial_j g_i$ is the element of DG in the i th row and j th column. Then for any $x, y \in D_0$ with $\|y\| = 1$, we have

$$\begin{aligned} \|DG(x)y\| &\leq C_1 \|DG(x)y\|_\infty \leq C_1 \|DG(x)\|_\infty \|y\|_\infty \\ &\leq C_1 C_2 \|y\| \max_i \sum_{j=1}^n |DG_{ij}(x)| \leq C_1 C_2 \max_i \sum_{j=1}^n m_{ij} = C < \infty, \end{aligned}$$

where the positive constants C_1 and C_2 are obtained from the equivalence of norms on $\mathbf{R}^n$. Thus, $\|DG(x)\| \leq C$ for all $x \in D_0$, which shows that DG is bounded in the induced norm, and $L < \infty$, so that G truly does satisfy a Lipschitz condition.

Problem 3.

Let $G : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be defined by

$$G(x) = \begin{bmatrix} \frac{1}{3} \cos(x_1 x_2) + \frac{1}{6} \\ -\frac{1}{20} e^{-x_1 x_2} - \frac{8}{20} \end{bmatrix}.$$

Set $x^{(0)} = 0$ and $D_0 = [-1, 1]^2$. Then if $x^{(k+1)} = G(x^{(k)})$ for $k \geq 0$, the sequence $\{x^{(k)}\}$ converges to a unique fixed point x^* of G in D_0 .

Proof. We begin by showing the $G(D_0) \subseteq D_0$. Indeed, if $x \in D_0$, then

$$\left| \frac{1}{3} \cos(x_1 x_2) + \frac{1}{6} \right| \leq \frac{1}{3} + \frac{1}{6} = \frac{1}{2},$$

and

$$\left| -\frac{1}{20} e^{-x_1 x_2} - \frac{8}{20} \right| \leq \frac{1}{20} e + \frac{8}{20} \leq \frac{9}{20}$$

because $e^{-x_1 x_2}$ is clearly maximal in D_0 when $x_1 = x_2 = -1$. The above two facts imply that $G(x) \in D_0$. Thus, $G(D_0) \subseteq D_0$.

Next, we compute the Jacobian DG of G :

$$DG(x) = \begin{bmatrix} -\frac{x_2}{3} \sin(x_1 x_2) & -\frac{x_1}{3} \sin(x_1 x_2) \\ \frac{x_2}{20} e^{-x_1 x_2} & \frac{x_1}{20} e^{-x_1 x_2} \end{bmatrix}.$$

In the induced infinity norm, we have

$$\|DG(x)\|_\infty = \max \left\{ \frac{|x_1| + |x_2|}{3} |\sin(x_1 x_2)|, \frac{|x_1| + |x_2|}{20} e^{-x_1 x_2} \right\}.$$

If $x \in D_0$, then $e^{-x_2 x_2} \leq e \leq 4$, so we get

$$\|DG(x)\|_\infty \leq \max \left\{ \frac{|x_1| + |x_2|}{3}, \frac{|x_1| + |x_2|}{5} \right\} \leq \frac{2}{3}.$$

Since the component functions of G are continuously differentiable on the closed and bounded set D_0 , their derivatives are also bounded and hence satisfy the hypotheses of Problem 2. Since the above shows that $L = \sup_{x \in D_0} \|DG(x)\|_\infty \leq \frac{2}{3} < 1$, it follows that G satisfies all hypotheses of Problem 2. We therefore conclude that G is a contraction on D_0 . Then G also satisfies the hypotheses of Problem 1. (note that $x^{(0)} = 0 \in D_0$), and we can conclude that $\{x^{(x)}\}$ converges to a unique fixed point x^* of G on D_0 . $\square$

Problem 4.

Let $A = \{a_{ij}\}$ be the $n \times n$ tridiagonal matrix with $a_{ii} = 2$ and $a_{i(i-1)} = a_{i(i+1)} = -1$ for all i that make sense. Then the eigenvectors of A are given by $\{\xi^k\}$ and $\{\lambda_k\}$, where

$$\lambda_k = 4 \sin^2 \left(\frac{\pi}{2} x_k \right), \quad \xi_j^k = \sin(k\pi x_j), \quad k = 1, 2, \dots, n, \quad j = 1, 2, \dots, n,$$

where $x_k = \frac{k}{n+1}$ for $k = 1, 2, \dots, n$. Furthermore, the ℓ^2 condition number of A is roughly given by

$$\text{cond}_2(A) \approx \frac{4(n+1)^2}{\pi^2}$$

when n is large.

Proof. Suppose that $n = 1$. Then $A = [2]$, of which the only eigenvalue is 2 with corresponding eigenvector $[1]$. Since $\lambda_1 = 4 \sin^2 \left(\frac{\pi}{2} \cdot \frac{1}{2} \right) = 2$, and $\xi_1^1 = \sin \left(\pi \cdot \frac{1}{2} \right) = 1$, we see that the claim is correct.

Now suppose that $n \geq 2$. Then for $k = 1, \dots, n$ and $j = 1, \dots, n$,

$$\begin{aligned} [A\xi^k]_j &= \sum_{m=1}^n a_{jm} \xi_m^k = \begin{cases} a_{j(j-1)} \xi_{j-1}^k + a_{jj} \xi_j^k + a_{j(j+1)} \xi_{j+1}^k & 1 < j < n \\ a_{11} \xi_1^k + a_{12} \xi_2^k & j = 1 \\ a_{n(n-1)} \xi_{n-1}^k + a_{nn} \xi_n^k & j = n \end{cases} \\ &= \begin{cases} -\sin \left(\frac{k(j-1)\pi}{n+1} \right) + 2 \sin \left(\frac{kj\pi}{n+1} \right) - \sin \left(\frac{k(j+1)\pi}{n+1} \right) & 1 < j < n \\ 2 \sin \left(\frac{k\pi}{n+1} \right) - \sin \left(\frac{2k\pi}{n+1} \right) & j = 1 \\ -\sin \left(\frac{k(n-1)\pi}{n+1} \right) + 2 \sin \left(\frac{kn\pi}{n+1} \right) & j = n. \end{cases} \\ &= -\sin \left(\frac{k(j-1)\pi}{n+1} \right) + 2 \sin \left(\frac{kj\pi}{n+1} \right) - \sin \left(\frac{k(j+1)\pi}{n+1} \right) \end{aligned}$$

in any case because $-\sin \left(\frac{k(j-1)\pi}{n+1} \right) = 0$ if $j = 1$, and $\sin \left(\frac{k(j+1)\pi}{n+1} \right) = 0$ if $j = n$. Thus, by the addition

formula for sine and the double-angle identity for cosine,

$$\begin{aligned}
 [A\xi^k]_j &= -\sin\left(\frac{kj\pi}{n+1}\right)\cos\left(\frac{k\pi}{n+1}\right) + \cos\left(\frac{kj\pi}{n+1}\right)\sin\left(\frac{k\pi}{n+1}\right) + 2\sin\left(\frac{kj\pi}{n+1}\right) \\
 &\quad - \sin\left(\frac{kj\pi}{n+1}\right)\cos\left(\frac{k\pi}{n+1}\right) - \cos\left(\frac{kj\pi}{n+1}\right)\sin\left(\frac{k\pi}{n+1}\right) \\
 &= 2\left(1 - \cos\left(\frac{k\pi}{n+1}\right)\right)\sin\left(\frac{kj\pi}{n+1}\right) \\
 &= 4\sin^2\left(\frac{k\pi}{2(n+1)}\right)\sin\left(\frac{kj\pi}{n+1}\right) \\
 &= 4\sin^2\left(\frac{\pi}{2}x_k\right)\sin(k\pi x_j) \\
 &= [\lambda_k \xi^k]_j.
 \end{aligned}$$

Thus, for all $k = 1, \dots, n$, we have $A\xi^k = \lambda_k \xi^k$, so λ_k is an eigenvalue of A , and ξ^k is an eigenvector. Since the values $\frac{\pi}{2}x_k \in [0, \frac{\pi}{2}]$ are distinct for $k = 1, \dots, n$, and sine is one-to-one on this interval, it follows that the eigenvalues $\lambda_1, \dots, \lambda_n$ are distinct. Since the matrix A can have no more than n distinct eigenvalues, it follows that $\{\lambda_k\}$ and $\{\xi^k\}$ constitute all the eigenvalues and eigenvectors of A .

The matrix A is symmetric, and, because all its eigenvalues are positive, it is also positive definite (note that $\lambda_k \neq 0$ for any k because $x_k \neq \pi m$ for any integer m , which are the only roots of $\sin^2$).

Thus, the ℓ^2 condition number of A is given by

$$\text{cond}_2(A) = \frac{\max_k \{\lambda_k\}}{\min_k \{\lambda_k\}}.$$

As mentioned above, sine and consequently $\sin^2$ are monotonic on the interval containing $\frac{\pi}{2}x_k$ for any k . In fact, $\sin^2$ is increasing; since the sequence $\{x_k\}$ is also increasing in k , it follows that $\{\lambda_k\}$ is increasing in k , giving $\max_k \{\lambda_k\} = \lambda_n$, and $\min_k \{\lambda_k\} = \lambda_1$. Then

$$\text{cond}_2(A) = \frac{\sin^2\left(\frac{n\pi}{2(n+1)}\right)}{\sin^2\left(\frac{\pi}{2(n+1)}\right)}.$$

Clearly, as $n \rightarrow \infty$ the numerator approaches 1, but the denominator approaches 0, meaning that the condition number of A approaches infinity as $n \rightarrow \infty$. We can estimate the rate of blow-up by using the Taylor series for sine centered at 0 to approximate the denominator for large n . We have $\sin(x) = x + \mathcal{O}(x^3)$ as $x \rightarrow 0$, so $\sin^2(x) \approx x^2 + \mathcal{O}(x^4)$ as $x \rightarrow 0$, yielding

$$\text{cond}_2(A) \approx \frac{1}{\left(\frac{\pi}{2(n+1)}\right)^2 + \mathcal{O}(n^{-4})} \approx \frac{4(n+1)^2}{\pi^2} \frac{1}{1 + \mathcal{O}(n^{-2})} \approx \frac{4(n+1)^2}{\pi^2} (1 + \mathcal{O}(n^{-2})) \approx \frac{4(n+1)^2}{\pi^2}.$$

as $n \rightarrow \infty$, where we have expanded the first fraction using a geometric series. $\square$

Problem 5.

Let A be a symmetric positive definite $n \times n$ matrix, and let $b \in \mathbf{R}^n$ be a nonzero vector. In the conjugate gradient method with initial guess $x^{(0)} = 0$, assume that $r^{(k)} \neq 0$ for $k < n$. Then all of the following hold for $1 \leq k \leq n$.

- (a) If $X_k = \text{span}\{x^{(1)}, x^{(2)}, \dots, x^{(k)}\}$, and $R_k = \text{span}\{r^{(0)}, r^{(1)}, \dots, r^{(k-1)}\}$, and $V_k = \text{span}\{v^{(1)}, v^{(2)}, \dots, v^{(k)}\}$, then $X_k = R_k = V_k$.
- (b) The residuals are pairwise orthogonal: $(r^{(j)}, r^{(k)}) = 0$ if $j < k$.
- (c) The direction vectors are pairwise A -orthogonal: $(v^{(j)}, Av^{(k)}) = 0$ if $j < k$.

Proof. Note that $x^{(1)} = x^{(0)} + t_1 v^{(1)} = t_1 v^{(1)} = t_1 r^{(0)}$, so clearly $X_1 = R_1 = V_1$. Now suppose for the sake of induction that $X_k = R_k = V_k$ for some $1 \leq k < n$.

Since $x^{(k+1)} = x^{(k)} + t_{k+1} v^{(k+1)}$, it follows that $x^{(k+1)} \in V_{k+1}$ because $x^{(k)} \in V_k \subseteq V_{k+1}$ by the inductive hypothesis. Hence, $X_{k+1} \subseteq V_{k+1}$.

Furthermore, since $v^{(k+1)} = r^{(k)} + s_k v^{(k)}$, it follows that $v^{(k+1)} \in R_{k+1}$ because $v^{(k)} \in R_k \subseteq R_{k+1}$ by the inductive hypothesis. Hence, $V_{k+1} \subseteq R_{k+1}$.

We showed in class that $t_{k+1} = \frac{(r^{(k)}, r^{(k)})}{(v^{(k+1)}, Av^{(k+1)})}$; therefore, the assumption $r^{(k)} \neq 0$ implies that $t_{k+1} \neq 0$. Thus, $r^{(k)} = v^{(k+1)} - s_k v^{(k)} = t_{k+1}^{-1} (x^{(k+1)} - x^{(k)}) - s_k v^{(k)} \in X_{k+1}$ because $v^{(k)} \in X_k \subseteq X_{k+1}$ by the inductive hypothesis. Hence, $R_{k+1} \subseteq X_{k+1}$.

Then $X_{k+1} = R_{k+1} = V_{k+1}$. This completes the proof of (a) by induction.

Now we prove (b) and (c). First, we observe that

$$(r^{(0)}, r^{(1)}) = (r^{(0)}, r^{(0)} - t_1 Av^{(1)}) = (r^{(0)}, r^{(0)}) - \frac{(r^{(0)}, r^{(0)})}{(r^{(0)}, Ar^{(0)})} (r^{(0)}, Ar^{(0)}) = 0$$

by the definition of t_1 , and

$$\begin{aligned} (v^{(1)}, Av^{(2)}) &= (Av^{(1)}, r^{(1)} + s_1 v^{(1)}) = (t_1^{-1} (r^{(0)} - r^{(1)}), r^{(1)} + s_1 r^{(0)}) \\ &= t_1^{-1} [s_1 (r^{(0)}, r^{(0)}) - (r^{(1)}, r^{(1)})] = 0 \end{aligned}$$

by the above and by the definition of s_1 . Second, suppose for the sake of induction that, for some $1 \leq k < n-1$ we have $(r^{(j)}, r^{(k)}) = 0$ if $j < k$ and $(v^{(j)}, Av^{(k+1)}) = 0$ if $j < k+1$. Then if $j < k$, we have

$$(r^{(j)}, r^{(k+1)}) = (r^{(j)}, r^{(k)} - t_{k+1} Av^{(k+1)}) = -t_{k+1} (r^{(j)}, Av^{(k+1)})$$

by the inductive hypothesis. Then, since $r^{(j)} = v^{(j+1)} - s_j v^{(j)}$, we have

$$(r^{(j)}, r^{(k+1)}) = -t_{k+1} (v^{(j+1)} - s_j v^{(j)}, Av^{(k+1)}) = 0$$

by the inductive hypothesis and the fact that $j < k$ implies $j+1 < k+1$. Additionally, we see that

$$(r^{(k)}, r^{(k+1)}) = (r^{(k)}, r^{(k)} - t_{k+1} Av^{(k+1)}) = (r^{(k)}, r^{(k)}) - \frac{(r^{(k)}, r^{(k)})}{(v^{(k+1)}, Av^{(k+1)})} (r^{(k)}, Av^{(k+1)}).$$

Since $r^{(k)} = v^{(k+1)} - s_k v^{(k)}$, we get

$$(r^{(k)}, Av^{(k+1)}) = (v^{(k+1)} - s_k v^{(k)}, Av^{(k+1)}) = (v^{(k+1)}, Av^{(k+1)})$$

by the inductive hypothesis. Thus,

$$(r^{(k)}, r^{(k+1)}) = 0.$$

Furthermore, if $j < k + 1$, then

$$\left(v^{(j)}, Av^{(k+2)}\right) = \left(v^{(j)}, Ar^{(k+1)} + s_{k+1}Av^{(k+1)}\right) = \left(Av^{(j)}, r^{(k+1)}\right)$$

by the inductive hypothesis as well as the fact that A is symmetric. Then, since $Av^{(j)} = t_j^{-1}(r^{(j-1)} - r^{(j)})$, we have

$$\left(v^{(j)}, Av^{(k+2)}\right) = t_j^{-1} \left(r^{(j-1)} - r^{(j)}, r^{(k+1)}\right) = 0$$

by the above and the fact that $j < k + 1$ and $j - 1 < k + 1$. Lastly, we see that

$$\begin{aligned} \left(v^{(k+1)}, Av^{(k+2)}\right) &= \left(Av^{(k+1)}, r^{(k+1)} + s_{k+1}v^{(k+1)}\right) = \left(t_{k+1}^{-1} \left(r^{(k)} - r^{(k+1)}\right), r^{(k+1)} + s_{k+1}v^{(k+1)}\right) \\ &= t_{k+1}^{-1} \left[s_{k+1} \left(r^{(k)} - r^{(k+1)}, v^{(k+1)}\right) - \left(r^{(k+1)}, r^{(k+1)}\right)\right]. \end{aligned}$$

By (a), we can choose $c_0, c_1, \dots, c_{k-1} \in \mathbf{R}$ such that $v^{(k+1)} = c_0r^{(0)} + c_1r^{(1)} + \dots + r^{(k)}$, where the last coefficient must be 1 because $v^{(k+1)} = r^{(k)} + s_kv^{(k)}$, and $v^{(k)} \in R_k$ by (a). Thus, applying pairwise orthogonality of the residuals from the inductive hypothesis and what we have proved for $r^{(k+1)}$, we have

$$\left(r^{(k)} - r^{(k+1)}, v^{(k+1)}\right) = \left(r^{(k)} - r^{(k+1)}, c_0r^{(0)} + c_1r^{(1)} + \dots + r^{(k)}\right) = \left(r^{(k)}, r^{(k)}\right).$$

Hence,

$$\left(v^{(k+1)}, Av^{(k+2)}\right) = t_{k+1}^{-1} \left[\frac{\left(r^{(k+1)}, r^{(k+1)}\right)}{\left(r^{(k)}, r^{(k)}\right)} \left(r^{(k)}, r^{(k)}\right) - \left(r^{(k+1)}, r^{(k+1)}\right)\right] = 0.$$

Then, by induction, given $1 \leq k \leq n-1$, we have $(r^{(j)}, r^{(k)}) = 0$ if $j < k$, and $(v^{(j)}, Av^{(k+1)}) = 0$ if $j < k+1$. This proves (c) because (c) is trivially true for $k = 1$; to prove (b), it remains to show that $(r^{(j)}, r^{(n)}) = 0$ if $j < n$. We do this using the same argument as above: if $j < n-1$, we have

$$\left(r^{(j)}, r^{(n)}\right) = \left(r^{(j)}, r^{(n-1)} - t_n Av^{(n)}\right) = -t_n \left(r^{(j)}, Av^{(n)}\right)$$

because we have proved (b) up to $k = n-1$. Then, since $r^{(j)} = v^{(j+1)} - s_j v^{(j)}$, we have

$$\left(r^{(j)}, r^{(n)}\right) = -t_n \left(v^{(j+1)} - s_j v^{(j)}, Av^{(n)}\right) = 0$$

by (c). Additionally, we see that

$$\left(r^{(n-1)}, r^{(n)}\right) = \left(r^{(n-1)}, r^{(n-1)} - t_n Av^{(n)}\right) = \left(r^{(n-1)}, r^{(n-1)}\right) - \frac{\left(r^{(n-1)}, r^{(n-1)}\right)}{\left(v^{(n)}, Av^{(n)}\right)} \left(r^{(n-1)}, Av^{(n)}\right).$$

Since $r^{(n-1)} = v^{(n)} - s_{n-1}v^{(n-1)}$, we get

$$\left(r^{(n-1)}, Av^{(n)}\right) = \left(v^{(n)} - s_{n-1}v^{(n-1)}, Av^{(n)}\right) = \left(v^{(n)}, Av^{(n)}\right)$$

by (c). Thus,

$$\left(r^{(n-1)}, r^{(n)}\right) = 0.$$

This completes the proof of (b). □

Problem 6.

Let $A \in \mathbf{R}^{n \times n}$ be symmetric positive definite, and $b \in \mathbf{R}^n$. Let x^* be the unique solution of $Ax = b$. Suppose

that the set $\sigma(A)$ of eigenvalues of A is the union of two subsets, so that $\sigma(A) = \sigma_0(A) \cup \sigma_1(A)$, and let $\ell = \#\sigma_0(A)$ be the number of elements in $\sigma_0(A)$. For $k \geq 0$, let $x^{(k)}$ be the k th iteration of the conjugate gradient method for solving $Ax = b$. Then, if $k \geq \ell$,

$$\|x^* - x^{(k)}\|_A \leq 2M \left(\frac{\sqrt{\kappa_1(A)} - 1}{\sqrt{\kappa_1(A)} + 1} \right)^{k-\ell} \|x^* - x^{(0)}\|_A,$$

where

$$\kappa_1(A) = \frac{\max(\sigma_1(A))}{\min(\sigma_1(A))}, \quad M = \max_{\lambda \in \sigma_1(A)} \prod_{\mu \in \sigma_0(A)} \left| 1 - \frac{\lambda}{\mu} \right|.$$

Proof. By the argument used to prove Theorem (VI) in class, we have

$$\|x^* - x^{(k)}\|_A^2 \leq \|x^* - x^{(0)}\|_A^2 \cdot \min_{q_k(0)=1} \max_{\lambda \in \sigma(A)} |q_k(\lambda)|^2$$

where q_k is a polynomial of degree at most k .

Now let $q_{k-\ell}$ be a given polynomial of degree at most $k - \ell$ (which is nonnegative because $k \geq \ell$) such that $q_{k-\ell}(0) = 1$. Then we can choose

$$q_k(x) = q_{k-\ell}(x) \prod_{\mu \in \sigma_0(A)} \left(1 - \frac{x}{\mu} \right).$$

It is clear that q_k has degree at most k and satisfies $q_k(0) = 1$. Furthermore, $q_k(\mu) = 0$ if $\mu \in \sigma_0(A)$. Thus,

$$\max_{\lambda \in \sigma(A)} |q_k(\lambda)| \leq \max_{\lambda \in \sigma_1(A)} |q_k(\lambda)| \leq \max_{\lambda \in \sigma_1(A)} |q_{k-\ell}(\lambda)| \cdot \max_{\lambda \in \sigma_1(A)} \prod_{\mu \in \sigma_0(A)} \left| 1 - \frac{\lambda}{\mu} \right| = M \cdot \max_{\lambda \in \sigma_1(A)} |q_{k-\ell}(\lambda)|.$$

Choosing $q_{k-\ell}$ the same way we did in the proof in class, that is,

$$q_{k-\ell}(x) = \frac{T_{k-\ell}\left(\frac{b-a-2x}{b-a}\right)}{T_{k-\ell}\left(\frac{b+a}{b-a}\right)},$$

where $T_{k-\ell}$ is the $(k - \ell)$ th Chebyshev polynomial, and $b = \max(\sigma_1(A))$ and $a = \min(\sigma_1(A))$, we can apply the same logic to obtain the inequality

$$|q_{k-\ell}(x)| \leq 2 \left(\frac{\sqrt{\kappa_1(A)} - 1}{\sqrt{\kappa_1(A)} + 1} \right)^{k-\ell},$$

where $\kappa_1(A) = \frac{b}{a}$. Thus,

$$\max_{\lambda \in \sigma(A)} |q_k(\lambda)| \leq 2M \left(\frac{\sqrt{\kappa_1(A)} - 1}{\sqrt{\kappa_1(A)} + 1} \right)^{k-\ell}.$$

It follows that

$$\min_{q_k(0)=1} \max_{\lambda \in \sigma(A)} |q_k(\lambda)| \leq 2M \left(\frac{\sqrt{\kappa_1(A)} - 1}{\sqrt{\kappa_1(A)} + 1} \right)^{k-\ell},$$

and combining this with the first inequality proves the theorem. $\square$

Problem 7.

Consider the boundary value problem

$$\phi'' = \kappa^2 \sinh(\phi) + f(x), \quad x \in (0, 4\pi), \quad \phi(0) = 0, \quad \phi(4\pi) = 0, \quad (5)$$

where $\kappa \in \mathbf{R}$ is a parameter, and $f(x) = -\sin(x) - \kappa^2 \sinh(\sin(x))$. The exact solution is $\phi(x) = \sin(x)$. Using a centered difference formula to approximate ϕ'' at the mesh points $\{x_i\}$, where $x_i = hi$ for $0 \leq i \leq N$, and $h = \frac{4\pi}{N}$ leads to the nonlinear system of equations

$$0 = F(\Phi) = -\frac{1}{h^2} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix} \begin{bmatrix} \Phi_1 \\ \vdots \\ \Phi_{N-1} \end{bmatrix} - \kappa^2 \begin{bmatrix} \sinh(\Phi_1) \\ \vdots \\ \sinh(\Phi_{N-1}) \end{bmatrix} - \begin{bmatrix} f(x_1) \\ \vdots \\ f(x_{N-1}) \end{bmatrix}, \quad (6)$$

where $\Phi = (\Phi_1, \dots, \Phi_{N-1})^T$ is an approximation of $(\phi(x_1), \dots, \phi(x_{N-1}))^T$.

- (a) We calculate the Jacobian of F by computing the value in its i th row and j th column, which is

$$DF_{ij}(\Phi) = \partial_j F_i(\Phi) = -\frac{1}{h^2} A_{ij} - \kappa^2 I_{ij} \cosh(\Phi_i),$$

where $F_i(\Phi)$ is the i th element of $F(\Phi)$, ∂_j is the partial derivative with respect to the j th component, I_{ij} is the element of the identity matrix in row i and column j , and $A \in \mathbf{R}^{(N-1) \times (N-1)}$ is given by

$$A = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}.$$

Then we can write $DF(\Phi)$ as

$$DF(\Phi) = \begin{bmatrix} -\frac{2}{h^2} - \kappa^2 \cosh(\Phi_1) & \frac{1}{h^2} & & & \\ \frac{1}{h^2} & -\frac{2}{h^2} - \kappa^2 \cosh(\Phi_2) & \frac{1}{h^2} & & \\ & \ddots & \ddots & \ddots & \\ & & \frac{1}{h^2} & -\frac{2}{h^2} - \kappa^2 \cosh(\Phi_{N-2}) & \frac{1}{h^2} \\ & & & \frac{1}{h^2} & -\frac{2}{h^2} - \kappa^2 \cosh(\Phi_{N-1}) \end{bmatrix}$$

- (b) The errors e_N are reported in Table 1. The code used to generate these values is in `p7b_output.txt`.
- (c) The average iterations are reported in Table 2. The code used to generate these values is in `p7c.m` and the outputs from running this script are in `p7c_output.txt`. Note that a tolerance of $\varepsilon = 10^{-10}$ is used for the forward error stopping condition in the conjugate gradient method.

N	e_N
32	5.270610e-03
64	1.319870e-03
128	3.301004e-04
256	8.253332e-05
512	2.063384e-05
1024	5.158493e-06

Table 1: Numerical errors using Newton's method to solve (6) with different values of N .

N	κ		
	1	10	100
64	8.25	8.00	7.20
1024	108.00	64.60	12.80

Table 2: $Aiters_{N,\kappa}$